ISEA INTERNSHIP

Assignment: Reliable UDP Communication in Mininet

Name : Piyush Kumar ,
Roll : CSB24051

Timeout used

0.5 s (roll number last digit is 1)

Mininet Topology

```
      s1
     /  \
    /    \
  h1      h2
```

h1 = UDP Server

h2 = UDP Client

s1 = OpenFlow Switch

Result table

Roll No	Name	Loss Percent	Timeout	Total Messages	Total Packets Sent	Total Retransmissions	Transfer Time (Seconds)	Status
CSB24051	Piyush Kumar	0	0.5	10	10	0	1.2401	SUCCESS
CSB24051	Piyush Kumar	5	0.5	10	10	0	1.3682	SUCCESS
CSB24051	Piyush Kumar	10	0.5	10	14	4	3.3719	SUCCESS

Short Answers

1. Why is ACK needed?

Ans: ACK (Acknowledgement) is used to confirm that a packet has been successfully received by the receiver.

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2. Why is timeout needed?

Answer: Timeout helps the sender detect packet or ACK loss. If no ACK is received within the timeout period, the packet is retransmitted.

3. Why can duplicate messages occur?

Answer: Duplicate messages occur when a packet is received successfully but its ACK is lost. The sender retransmits the packet, causing the receiver to receive it again.

4. What happens when packet loss increases?

Answer: As packet loss increases, retransmissions increase, transfer time increases, and network efficiency decreases.

5. How is this method similar to TCP?

Answer: Like TCP, it uses sequence numbers, acknowledgements (ACKs), timeouts, and retransmissions to provide reliable data delivery.

6. How is it different from TCP?

Answer: This method implements only basic Stop-and-Wait reliability, whereas TCP also provides connection establishment, flow control, congestion control, and sliding window mechanisms.